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ECE 20007

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# ECE 20007 Final Project Report - Audio Equalizer

## Intro / Abstract:

### 1.2: Definition and Purpose of Project

This project focused on designing, simulating, and validating an analog audio equalizer with three independent frequency bands (bass, mid, treble) using LM324 operational amplifiers and an LM386 power amplifier. An audio equalizer is a signal processing circuit designed to isolate and manipulate specific frequency components of an audio signal. The purpose of this project is to apply filtering, amplification, and signal summation together to build a device that can take a standard 320mV RMS audio input and allow for user-controlled frequency adjustments. The primary objectives are to split the signal into Bass, Mid, and Treble channels using RC filter networks, recombine them through an active summation amplifying stage to keep a stable 100mV RMS signal, $V_{amp}$, and provide sufficient current through a power amplifier to drive a speaker with at least 400mW of power.

### 1.3: Key Functions of the Audio Equalizer

The audio equalizer has four main functions that work together in the system. First is filtering, which separates the input into specific frequency bands. Second is the gain adjustment, which allows for attenuation of the signal via potentiometers. Third is summation, where the processed signals are mixed back into a single output without interfering with one another. Fourth is buffering, which uses the op-amps to isolate stages of the circuit. This isolation makes sure that the output of one stage does not interfere with the input of the next, which keeps the frequency cutoffs stable and accurate.

### 1.4: Applications of Audio Equalizers

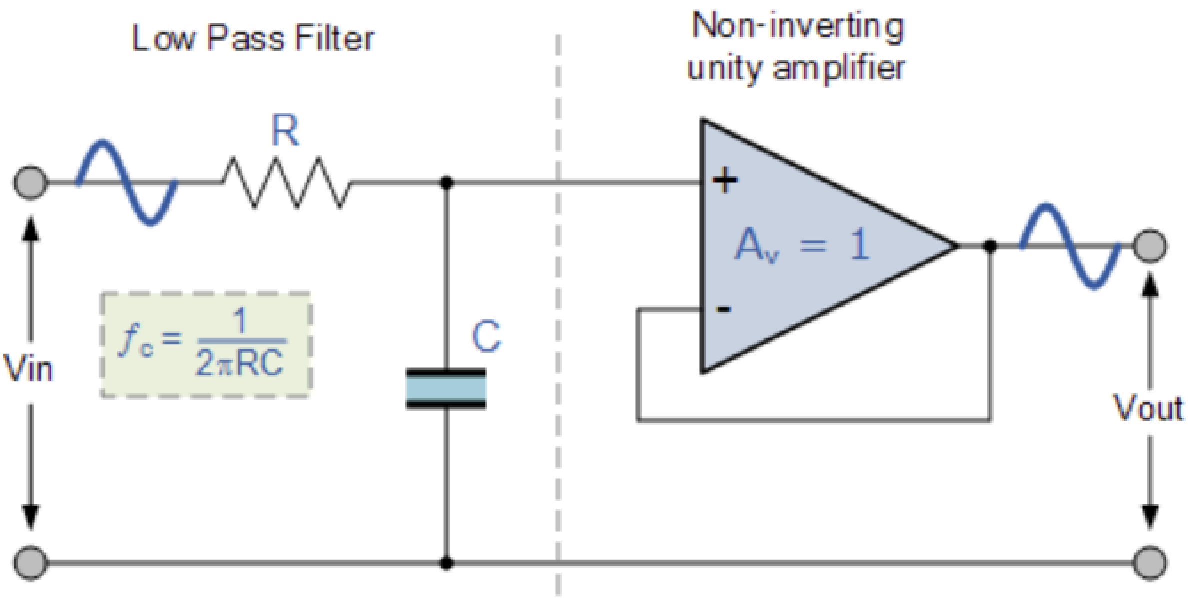
Audio equalizers are used in a wide range of practical applications across the electronics industry. In consumer electronics like smartphones, car radios, and home theater systems, they allow users to adjust audio to account for their room acoustics or personal preference in general. In professional audio production, equalizers are useful for cleaning up recordings by removing unwanted frequencies, and help prevent different instruments from interfering with each other in the same frequency range.

## Theory:

### 2.1: General Filter Working Principles

The reason why the audio equalizer system is able to separate frequencies is achieved through first-order active RC filters. These filters use a passive RC network to define the frequency response, which is then followed by a non-inverting buffer to provide isolation between stages. The main relationship which controls the filter is capacitive reactance ($X_C$) which is defined as: $X_C = \frac{1}{2\pi f C}$

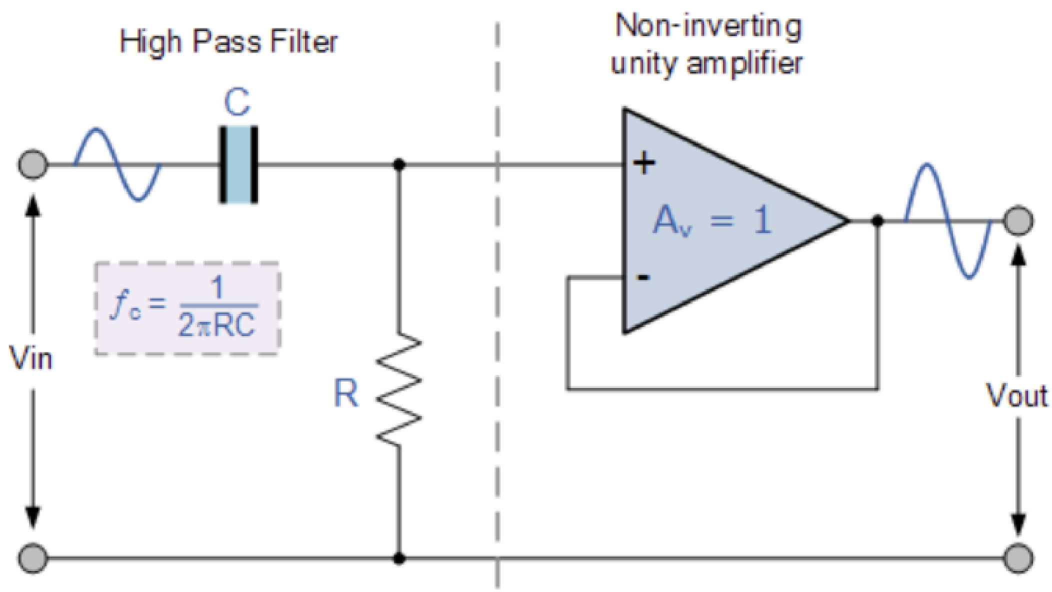
**A. *Active Low Pass Filter Overview***




The low pass filter is designed to pass signals below a specific cutoff frequency while attenuating higher frequencies. In this configuration, the resistor is placed in series with the input, and the capacitor is placed in parallel with the load. As the frequency increases, the capacitive reactance decreases, which ends up shorting high-frequency components to ground. The buffer stage then recreates this filtered voltage at its output.

The -3dB cutoff frequency is calculated by: $f_{c,low} = \frac{1}{2\pi RC}$

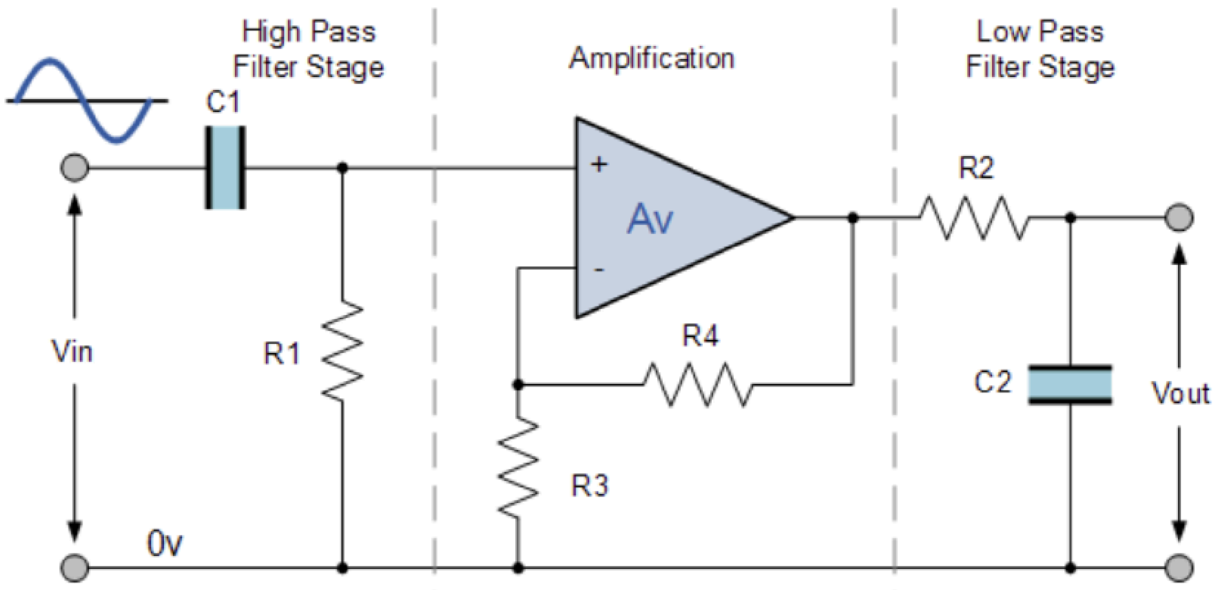
**B.** ***Active High Pass Filter Overview***




The high pass filter passes signals above the cutoff frequency and blocks lower frequencies and DC components. In this circuit, the capacitor is placed in series with the signal path, and the resistor is in parallel with the load. At low frequencies, the capacitive reactance is very high, which acts as an open circuit. As the frequency increases, $X_C$ drops, which allows the signal to pass through to the buffer stage.

The -3dB cutoff frequency is calculated by: $f_{c,high} = \frac{1}{2\pi RC}$

**B.** ***Active Mid Pass (Band Pass) Filter Overview***

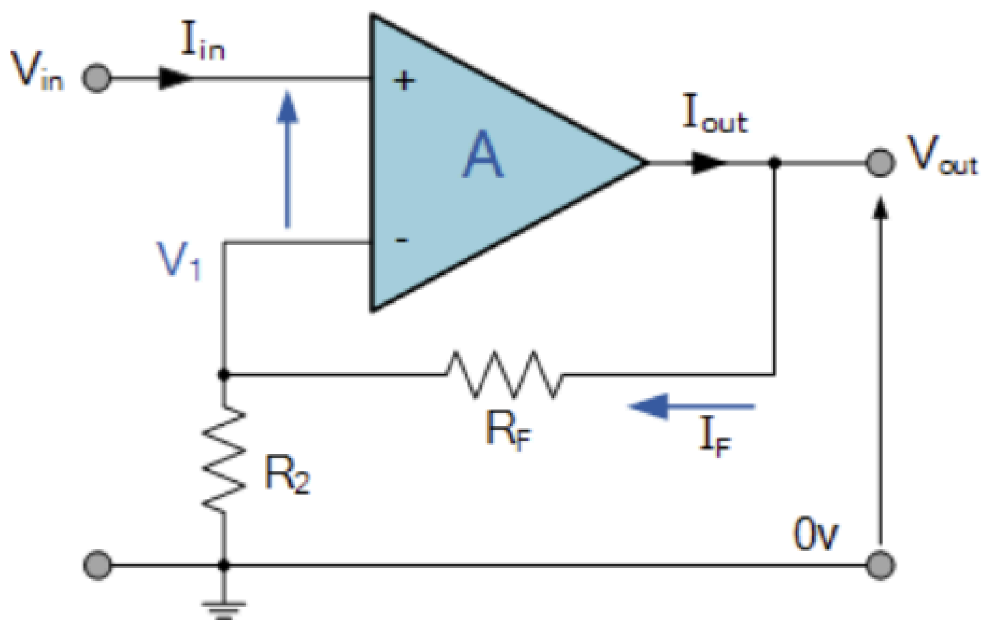



The mid pass filter is implemented by cascading a high pass and low pass stage around an amplification stage. This configuration creates a passband, which allows only a certain range of middle frequencies to reach the output. The first stage (C1 and R1) acts as a high pass filter to set the lower cutoff frequency. The signal is then amplified through an op-amp, which is then sent through a low-pass stage (R2 and C2), which sets the upper cutoff frequency. The resulting output is a frequency response where the bandwidth is between two independent -3dB cutoff frequencies. The -3dB cutoff frequency is calculated by using the same equations for low and high pass listed before, and the bandwidth / total range of frequencies passed by the filter can be calculated as : $BW = f_{c,\,high} - f_{c,\,low}$

**2.2: Operational Amplifier Working Principles**

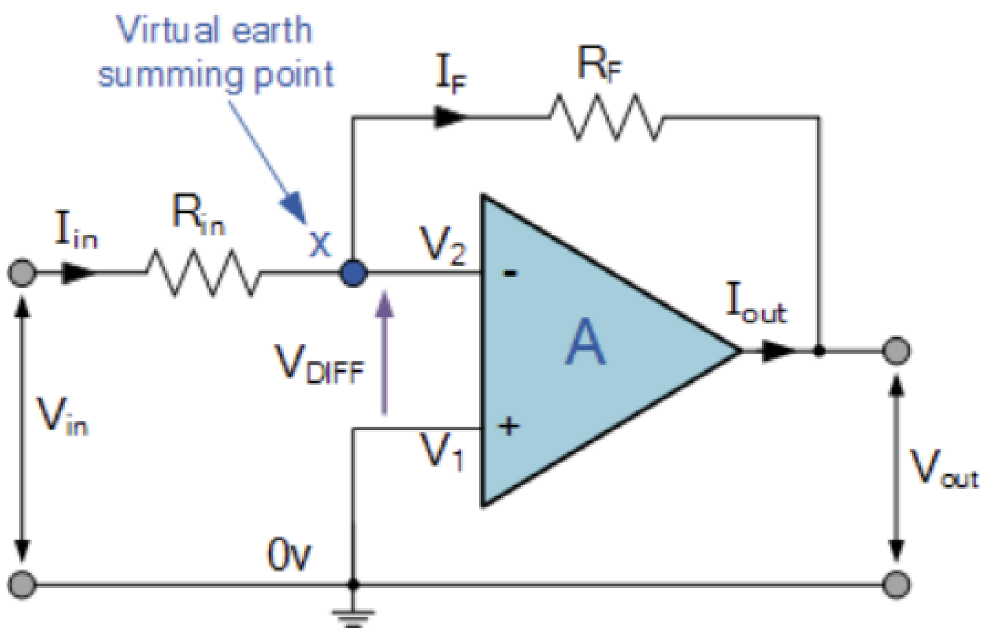
An operational amplifier, or op-amp, is a component used to amplify, buffer, or combine signals. In this project, the LM324 op-amp was used to manage signal levels and ensure that different parts of the circuit don't interfere with each other. It was set up in a negative feedback configuration to ensure stable signal processing by feeding the output back to the inverting input, the op-amp maintains a short between its input terminals which allows for control over gain and prevents signal distortion.

**A.** ***Non-Inverting Buffer***




The non-inverting buffer is used immediately following the RC filter networks to provide isolation between stages. The buffer's configuration has a voltage gain of exactly one ($A_V = 1$). While a standard non-inverting amplifier's gain is defined by $A_v = 1 + \frac{R_F}{R_2}$. To get a buffer with a voltage gain of one, the feedback resistance ($R_f$) is set to zero ohms (a short) and the grounded resistor (R2) is set to infinity (an open circuit). The primary function is to ensure that the calculated frequency cutoffs from the passive RC networks remain accurate regardless of the next stages. This relationship is shown with the equation $V_{out} = V_{in}$

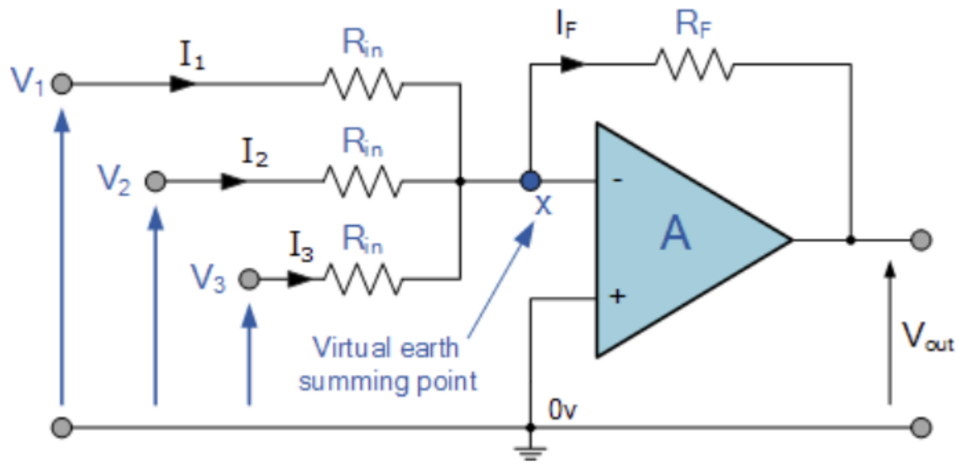
**B.** ***Inverting Amplifier***




The inverting amplifier is used in the volume control stage of each frequency band. In this configuration, the non-inverting (+) terminal is connected to ground, and then the output signal is sent through a feedback loop back to the inverting (-) terminal through a feedback resistor. The gain is determined by the ratio of the feedback resistor to the input resistor. By using a potentiometer for $R_f$, the gain can be manually adjusted to attenuate or boost the signal.

The equation for the output voltage is: $V_{out} = -Vin(\frac{R_f}{R_{in}})$

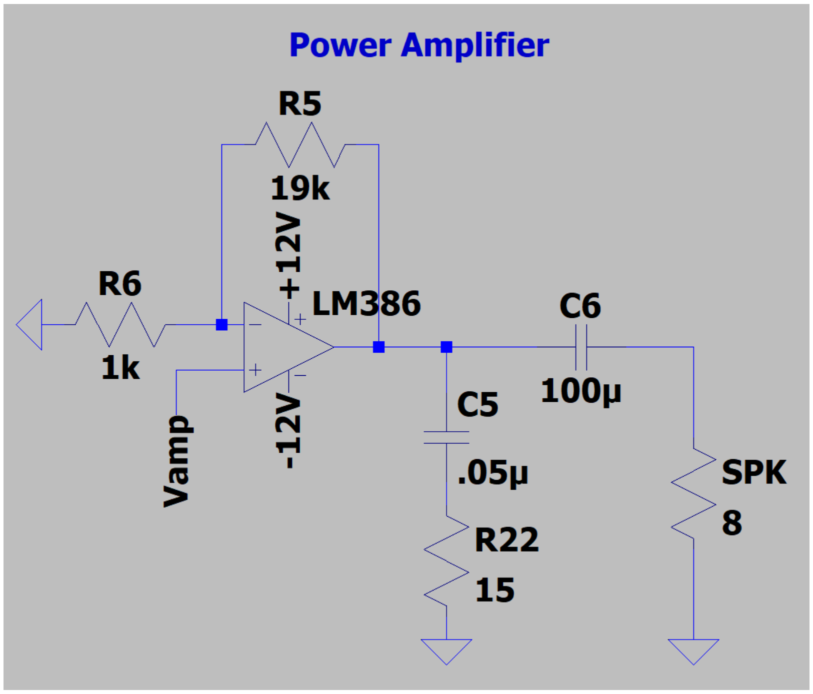
**C.** ***Summation Amplifier***




The summing amplifier serves as the recombination stage, which mixes the Bass, Mid, and Treble signals into a single output voltage, $V_{out}$. Because of the virtual ground at the inverting input, the multiple branches are added together linearly without interference. The final output is the inverted sum of the input voltages, which are weighted by the ratio of the feedback resistor and each of their specific input resistors. In the project, this is the stage that is responsible for producing the $V_{amp}$ signal before it is sent to the power amplifier. This relationship is shown with this equation: $V_{out} = -R_f(\frac{V1}{R_{in}} + \frac{V2}{R_{in}} + \frac{V3}{R_{in}})$

**2.3: Power Amplifier Working Principles**

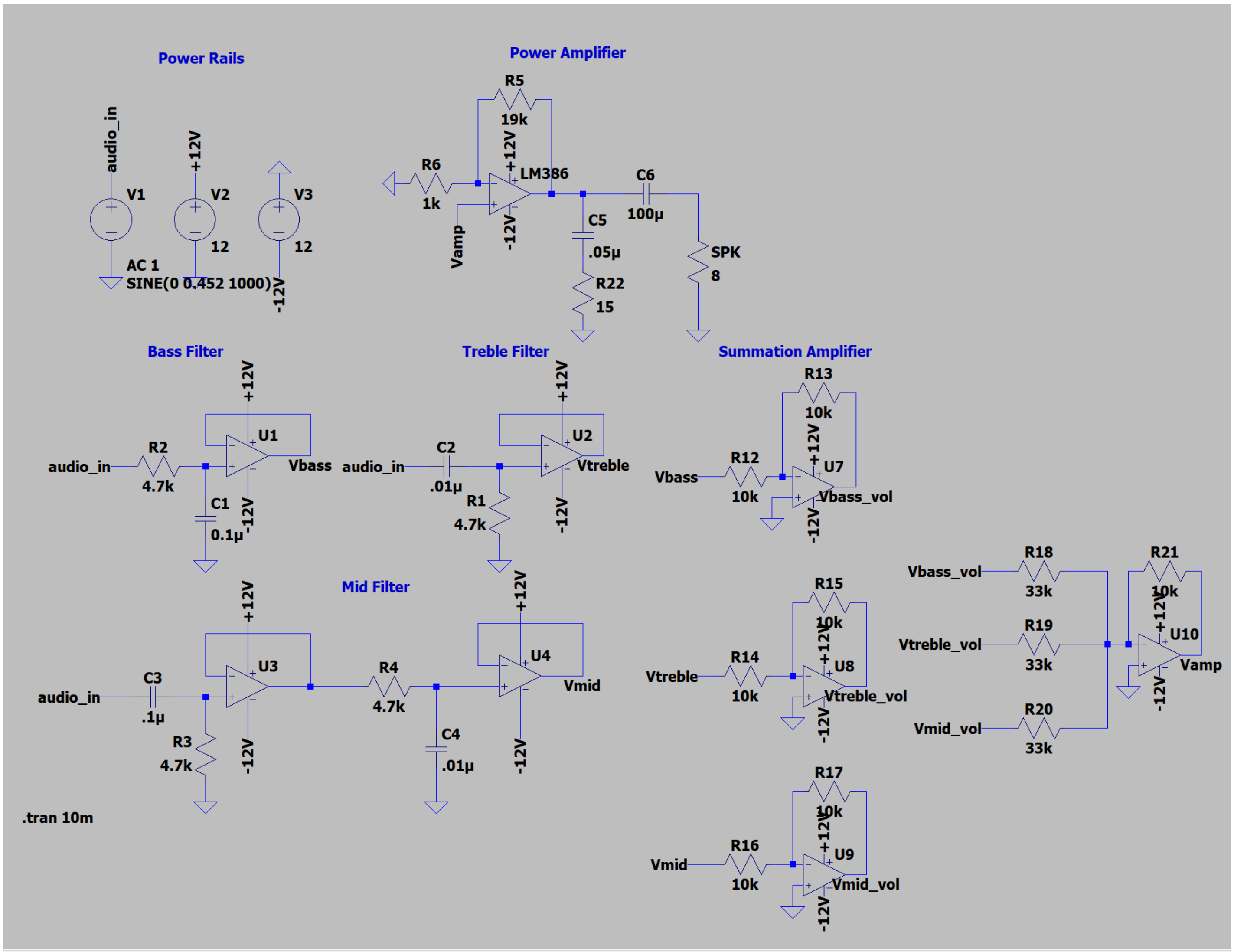
**C.** ***Power Amplifier***




The main working principle of a power amplifier is to take a processed voltage signal and provide the necessary current gain to drive a low impedance load, such as a speaker in the case of this project. While standard op-amps such as the LM324 focus on voltage amplification, they do not source enough current for the speaker, so instead the LM386 is used which draws high current from DC power supply and delivers it to the load in proportion to the input signal. The average output power is related by: $P_{out} = \frac{V_{rms}^{2}}{R_{load}}$ where $V_{rms}$ is the RMS voltage of $V_{out}$ and $R_{load}$ is the impedance of the speaker.

# Experimental Procedure:

## 3.1: Schematic of Audio Equalizer



## 3.2: Design Parameters

A. Filter design parameters (Low pass, high pass, mid pass)

Low-Pass Filter (Bass) Calculations:

| Component | Ideal Calculation | Actual Value Used |
|---|---|---|
| Capacitor | 0.1uF | 0.1uF |
| Resistor | 4,974 ohms | 4.7k + 470 = 5,170 ohms |
| Actual Calculated | 320 Hz | 308 Hz |
| Percent Error | 0% | 3.75% |

High-Pass Filter (Treble) Calculations:

| Component | Ideal Calculation | Actual Value Used |
|---|---|---|
| Capacitor | 0.01uF | 0.01uF |
| Resistor | 4,974 ohms | 4.7k + 560= 5,260 ohms |
| Actual Calculated | 3200 Hz | 3026 Hz |
| Percent Error | 0% | 5.44% |

Band-Pass Filter (Mid) Calculations:

High-Pass Stage (lower frequency cutoff):

| Component | Ideal Calculation | Actual Value Used |
|---|---|---|
| Capacitor | 0.1uF | 0.1uF |
| Resistor | 4,974 ohms | 4.7k + 470= 5,170 ohms |
| Actual Calculated | 320Hz | 308Hz |

Low-Pass Stage (higher frequency cutoff):

| Component | Ideal Calculation | Actual Value Used |
|---|---|---|
| Capacitor | 0.01uF | 0.01uF |
| Resistor | 4,974 ohms | 4.7k + 270= 4970 ohms |
| Actual Calculated | 3200Hz | 3202Hz |

B. Volume control circuits design parameters

The volume control stage uses the inverting amplifiers where gain is determined by the ratio of the feedback resistance to the input resistance.

Volume Control Calculations:

- Formula: $Gain\ =\ \frac{-R_f}{R_{in}}$
- Input Resistor ($R_{in}$): 10k Ohms
- Potentiometer ($R_f$): 10k Ohms (variable)
- Maximum Gain: $\frac{-10k\Omega}{10k\Omega} =-\ 1$
- Minimum Gain: $\frac{0\Omega}{10k\Omega} =\ 0$

C. Signal recombination stage design parameters

The summing amplifier combines the three signals using specific input resistors to scale the final output voltage to the target level.

Summing Amplifier Calculations:

- Formula: $V_{amp} = - R_f(\frac{V_{bass}}{R_1} + \frac{V_{mid}}{R_2} + \frac{V_{treble}}{R_3})$
- Input Resistors ($R_1$, $R_2$, $R_3$): 33k Ohms each
- Feedback Potentiometer ($R_f$): 10k Ohms
- Total Gain per channel: $\frac{-10k\Omega}{33k\Omega} = - 0.303$

Final Power Calculation:

- Target power: > 400mW
- Speaker Load: 8 Ohms
- Speaker Voltage ($V_{rms}$): 1.94Vrms
- Actual Power Output: $P = \frac{V_{rms}^2}{R} = \frac{1.94^2}{8} \approx 470mW$

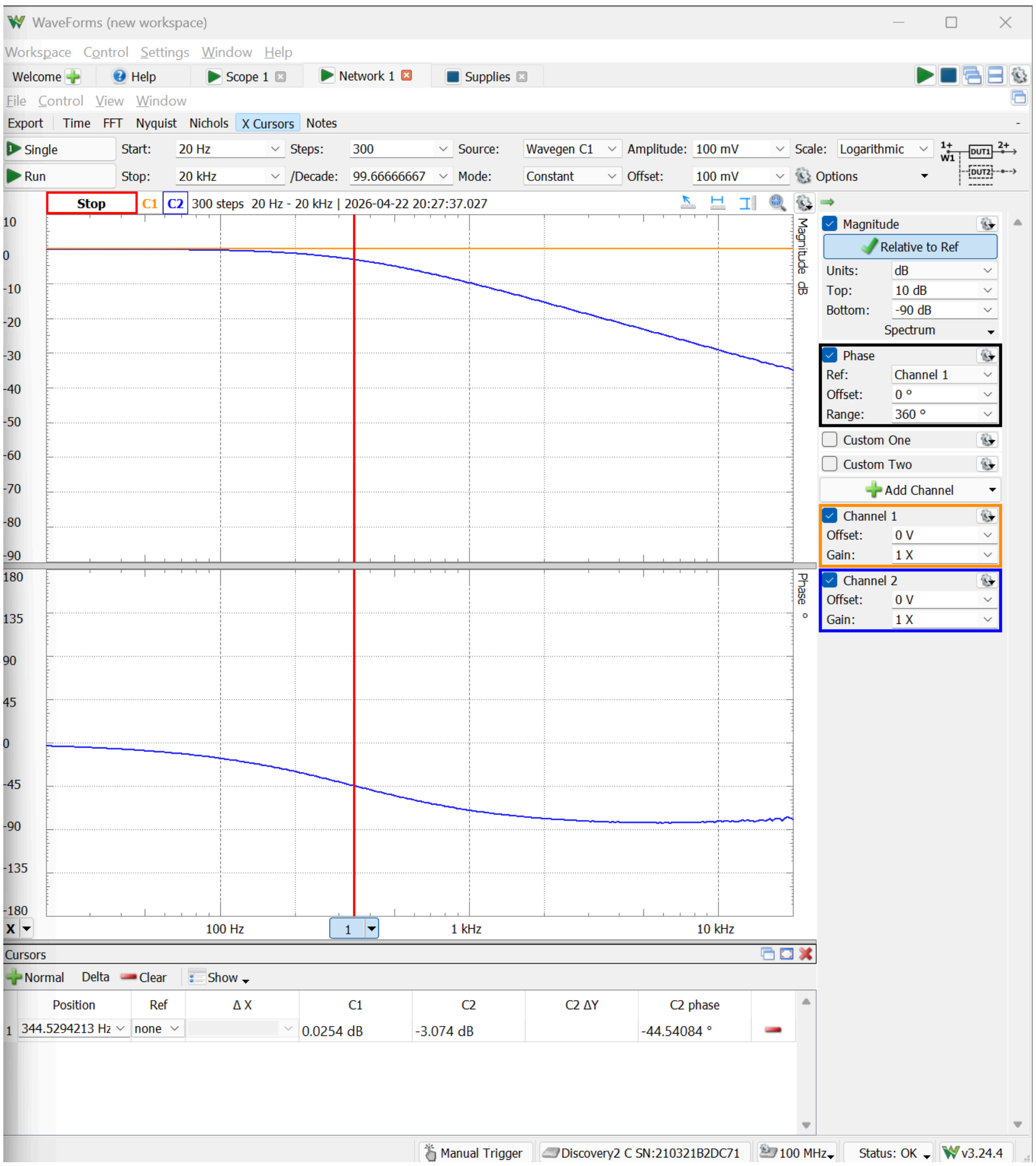
**3.3: Short Description of Circuit Operation**

The audio equalizer circuit operates by taking a full-range input audio signal and splitting it into three parallel processing paths. First, the signal passes through active RC filter networks: low pass (bass), a band pass filter (mid), and a high pass filter (treble), which all together isolate specific frequency bands using cutoffs at 320Hz and 3200Hz. Non-inverting buffers then follow the passive networks to prevent interference and signal loss between different stages of the circuit.

Next, the isolated signals enter the volume control stage, which uses inverting op amps with potentiometers. This allows the user to manually cut or boost the volume of each specific frequency band. These three independently adjusted signals are then routed into a summing amplifier, which combines them back into a single mixed audio signal. Finally, the combined signal is fed into an LM386 power amplifier, which provides at least 400mW of power to drive the physical speaker.

# Results:

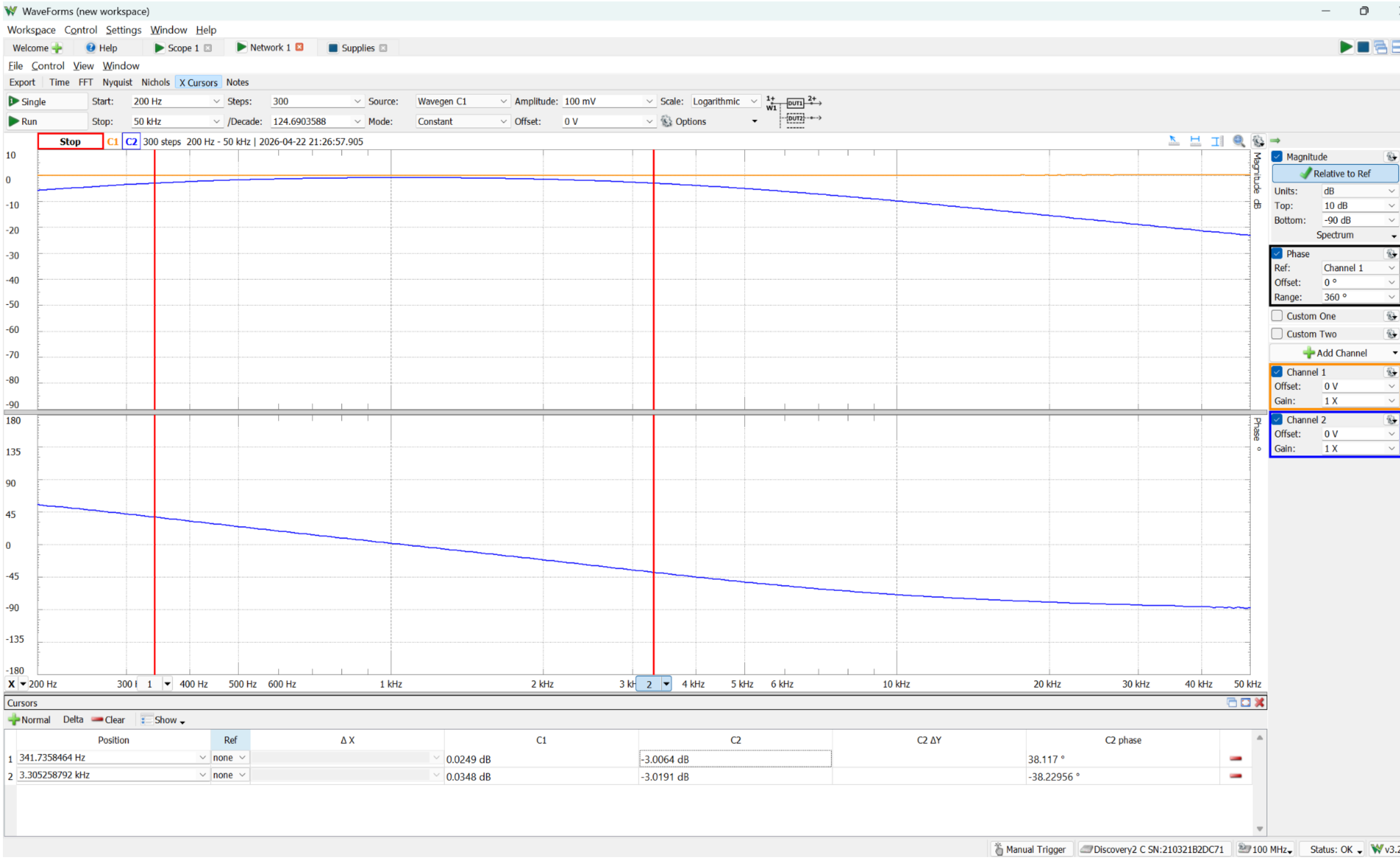
## 4.1 Lowpass filter's FRA curves with explanation



Explanation:

The frequency response of the low-pass filter (bass) was measured using the Analog Discovery 2, and cursor 2 was placed at the -3dB point which measured a cutoff frequency of 344.53Hz. Compared to the design target of 320Hz, this is a +7.66% error, which satisfies the 10% tolerance.

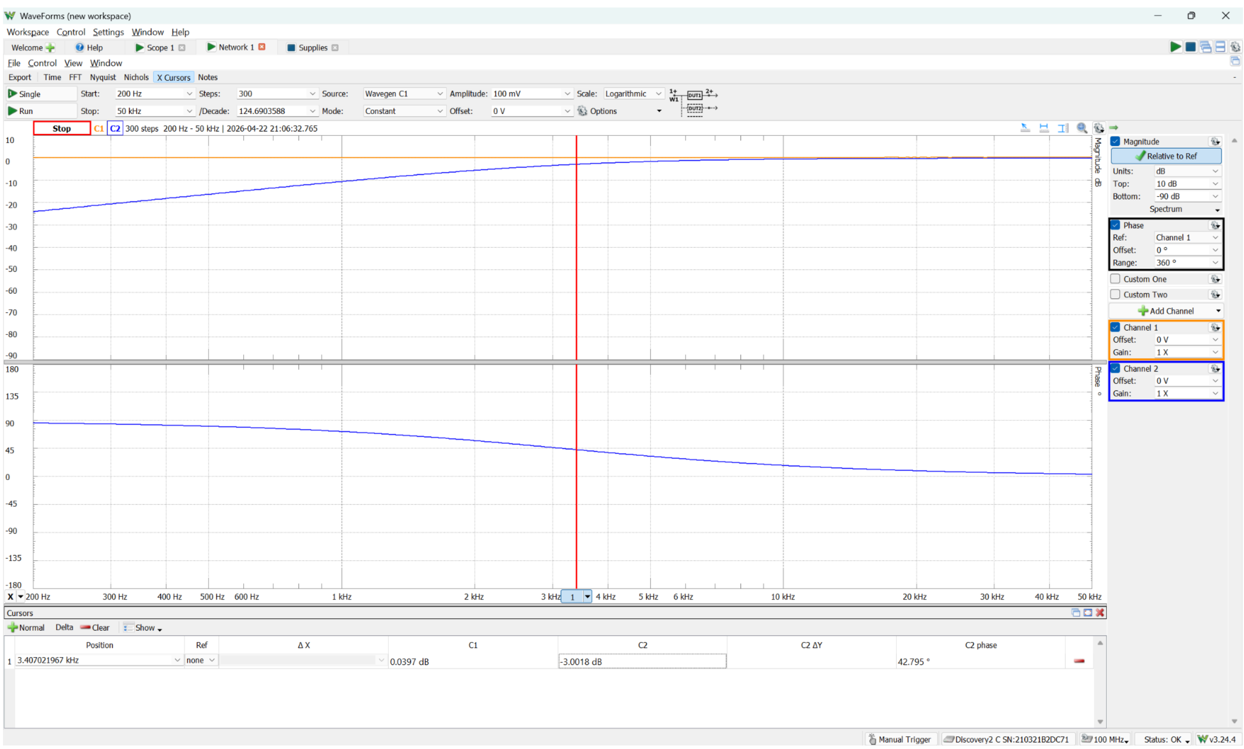
## 4.2 Mid-pass filter’s FRA curves with explanation




Explanation:

The frequency response of the mid-pass filter was also measured using the AD2 and verified that the passband region worked properly on the breadboard. The top curve clearly shows attenuation at both the low and high frequency cutoffs, with a flat passband in the middle frequencies. Two cursors were placed at their respective -3dB points to identify the physical cutoff frequencies. Cursor 1 marks the lower (high-pass) cutoff at 341.74 Hz, resulting in a +6.79% error compared to the 320 Hz target. Cursor 2 marks the upper (low-pass) cutoff at 3.305 kHz, resulting in a 3.29% error against the 3200 Hz target. As a result, both measured frequencies successfully fall within the 10% design tolerance

## 4.3 High pass filter’s FRA curves with explanation



Explanation:

The frequency response of the high-pass filter (treble) was also captured using the AD2. The magnitude response (top curve) properly exhibits the expected behavior of a high pass network, showing the attenuation of the low-frequency signals and a flat passband for frequencies above the cutoff point.

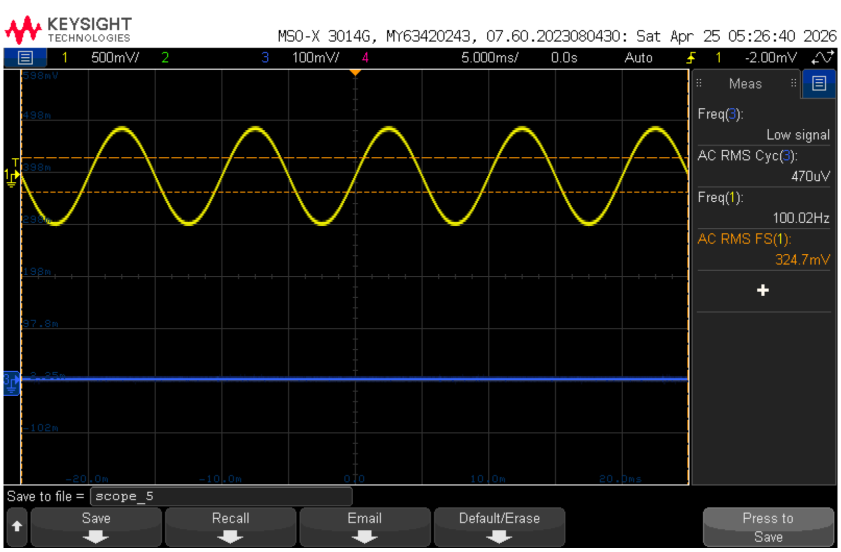
As shown in the image, a cursor was placed at the -3dB point to determine the physical cutoff frequency. The measured cutoff frequency was approximately 3.407 kHz, which when compared to the design target of 3200 Hz, yields a +6.47% error, which successfully meets the 10% project tolerance.

**4.4 Plots showing Vamp with all the equalizer turned to minimum setting with explanation**

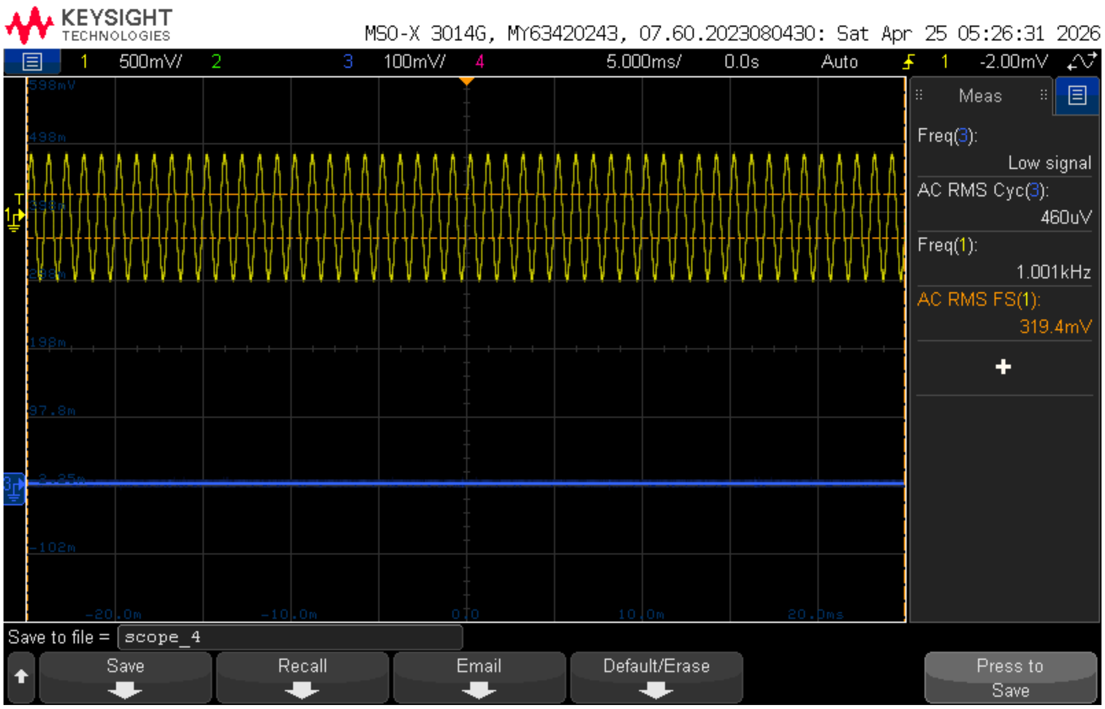
To verify the signal cut capabilities of the volume stage, the potentiometers for all three frequency bands were set to their minimum settings. A full-range input signal of approximately 320 mVrms (yellow trace on channel 1) was applied to the circuit, and the recombined output signal, $V_{amp}$ (blue trace on channel 3) was measured at 100 Hz, 1 kHz, and 10 kHz.

The design requirement states that at the minimum volume setting, the output signal $V_{amp}$ must be less than 15mVrms. As shown in the oscilloscope captures, the output traces are completely flat. The measured AC RMS voltages are in the microvolt range or below the detection threshold of the oscilloscope, which successfully satisfies that requirement.

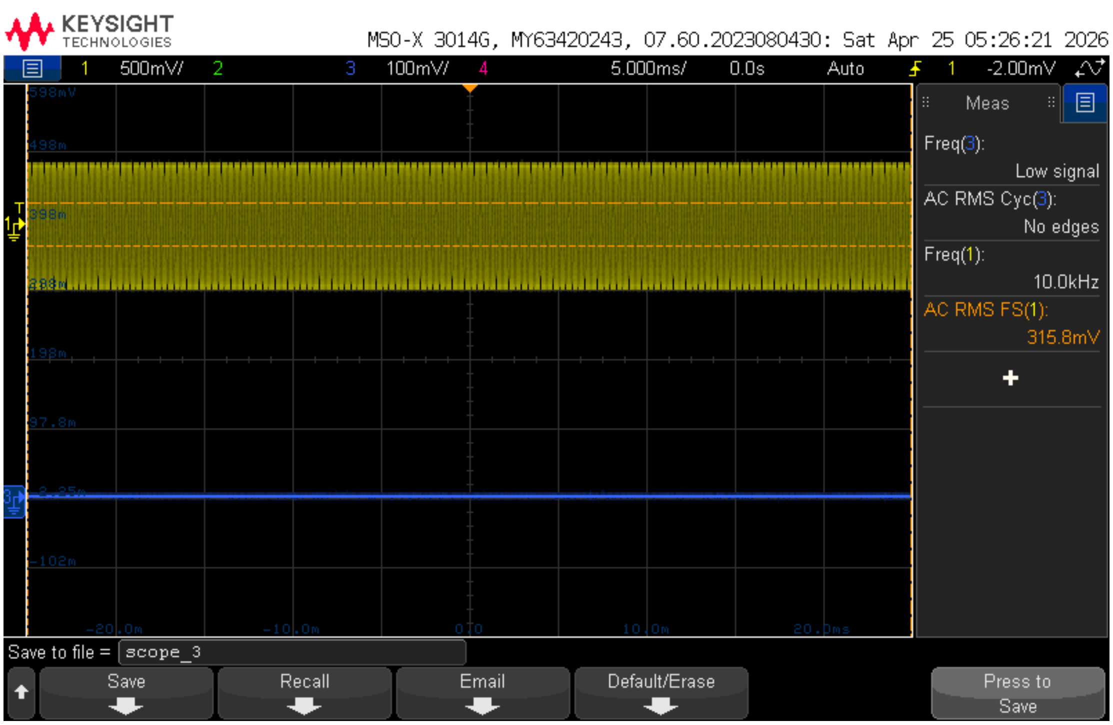
100 Hz measurement:



1 kHz measurement:




10 kHz measurement:



Explanation:

| Test Frequency | Input Voltage ($V_{in}$) | Output Voltage ($V_{amp}$) | Satisfies <15 mVrms Requirement |
|---|---|---|---|
| 100 Hz (bass) | 324.7 mVrms | 470 uVrms (0.47 mVrms) | Yes |
| 1 kHz (mid) | 319.4 mVrms | 460 uVrms (0.46 mVrms) | Yes |
| 10 kHz (treble) | 315.8 mVrms | ~0 mVrms (no edges detected) | Yes |

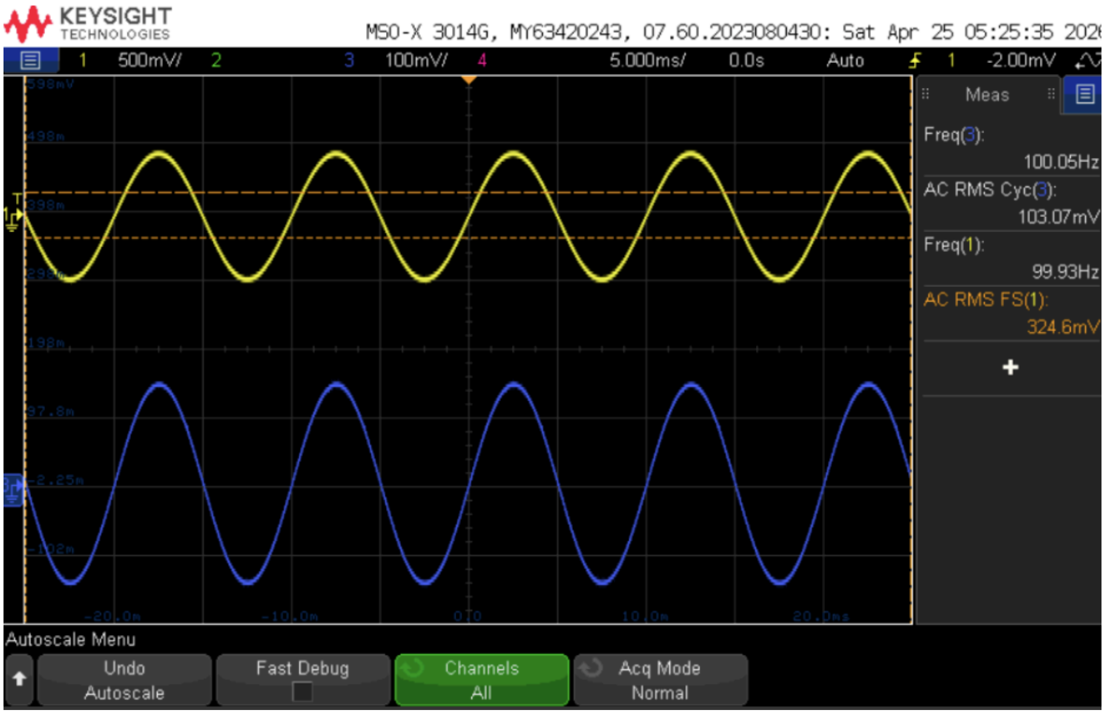
The data confirms that the combination of the inverting volume control amplifiers and the summing amplifier successfully scales the 320 mVrms input down to the target 100mVrms level across all three channels at the maximum equalizer setting.

**4.5 Plots showing Vamp with all the equalizer turned to maximum setting with explanation**

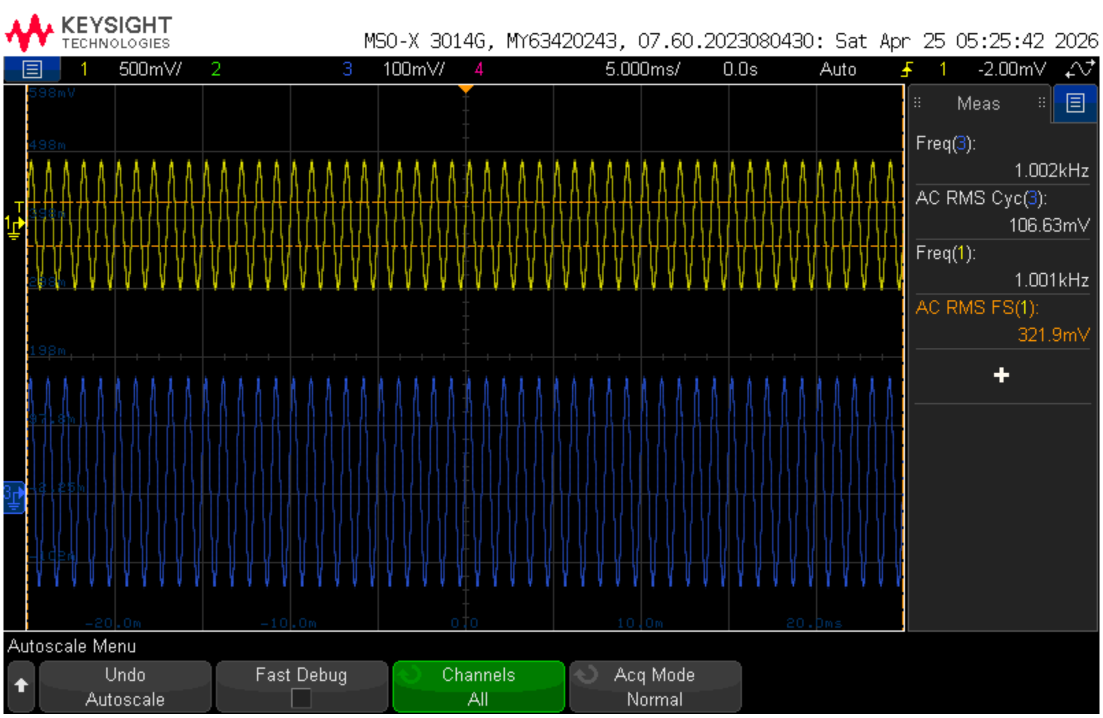
Similarly, to verify the maximum gain of the volume control and recombination stages, a full-range input signal of approximately 320 mVrms (yellow trace on channel 1) was applied to the circuit. All three potentiometers were turned to their maximum settings (10k Ohms) to allow the full signal through the respective filters. The mixed output signal, $V_{amp}$ (blue trace on channel 3), was then measured at three frequencies representing the bass, mid, and treble passbands: 100 Hz, 1 kHz, and 10 kHz.

The design requirement for the maximum volume output at $V_{amp}$ is 100 mVrms with a 10% tolerance, and as shown in the oscilloscope captures, the measured output voltages for all three frequencies successfully fall within this tolerance window.

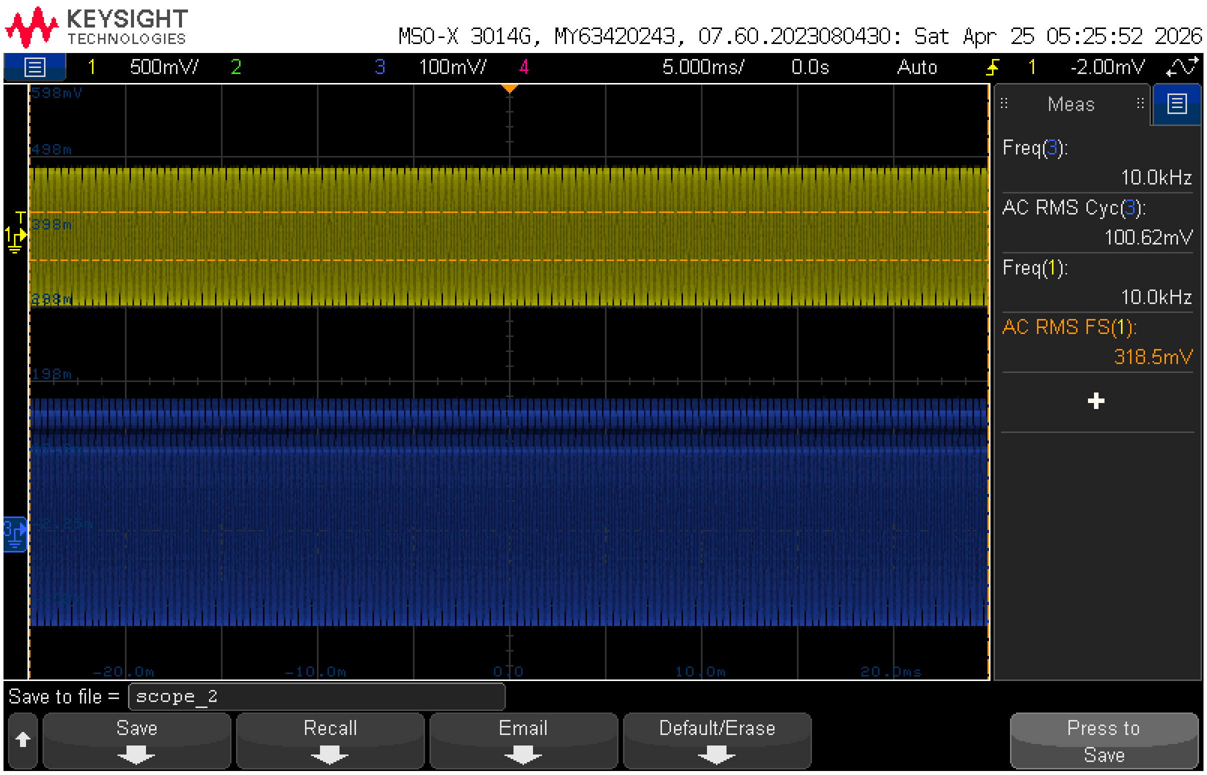
100 Hz measurement:




1 kHz measurement:



10 kHz measurement:



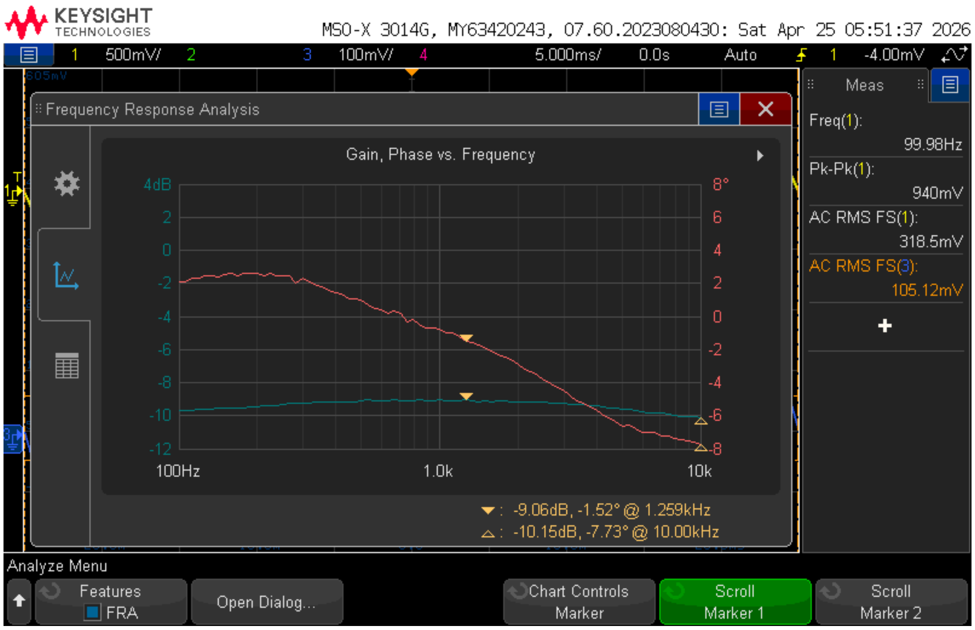

<u>Explanation:</u>

| Test Frequency | Input Voltage ($V_{in}$) | Output Voltage ($V_{amp}$) | Percent Error From 100mVrms Target |
|---|---|---|---|
| 100 Hz (bass) | 324.6 mVrms | 103.07 mVrms | +3.07% |
| 1 kHz (mid) | 321.9 mVrms | 106.63 mVrms | +6.63% |
| 10 kHz (treble) | 318.5 mVrms | 100.62 mVrms | +0.62% |

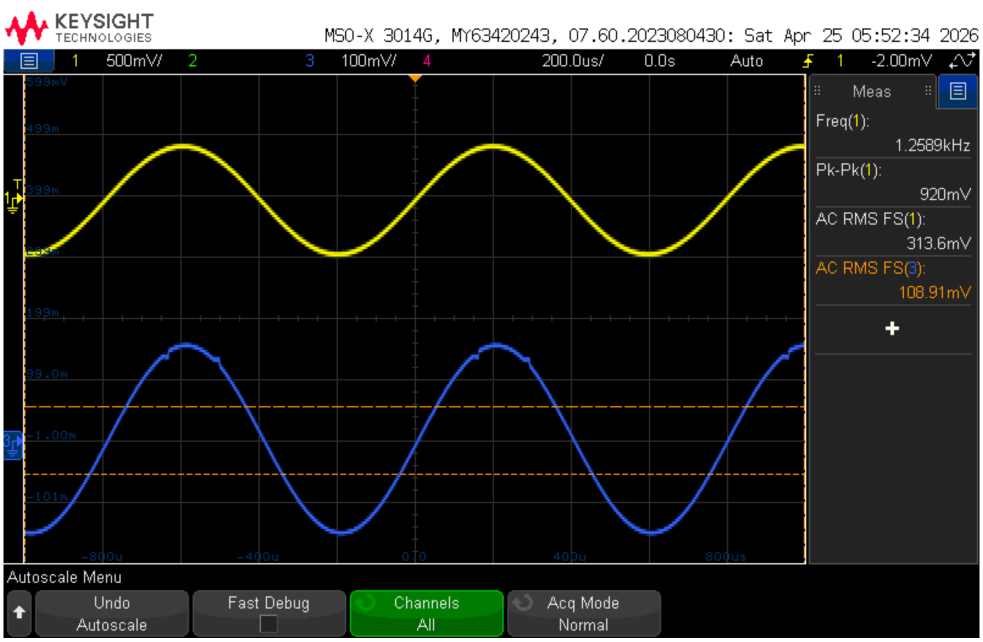
The data confirms that the combination of the inverting volume control amplifiers and the summing amplifier successfully scales the 320 mVrms input down to the target 100mVrms level across all three channels at the maximum equalizer setting.

## 4.6 Plots related to Ripple measurement (15 mVrms ± 10%)

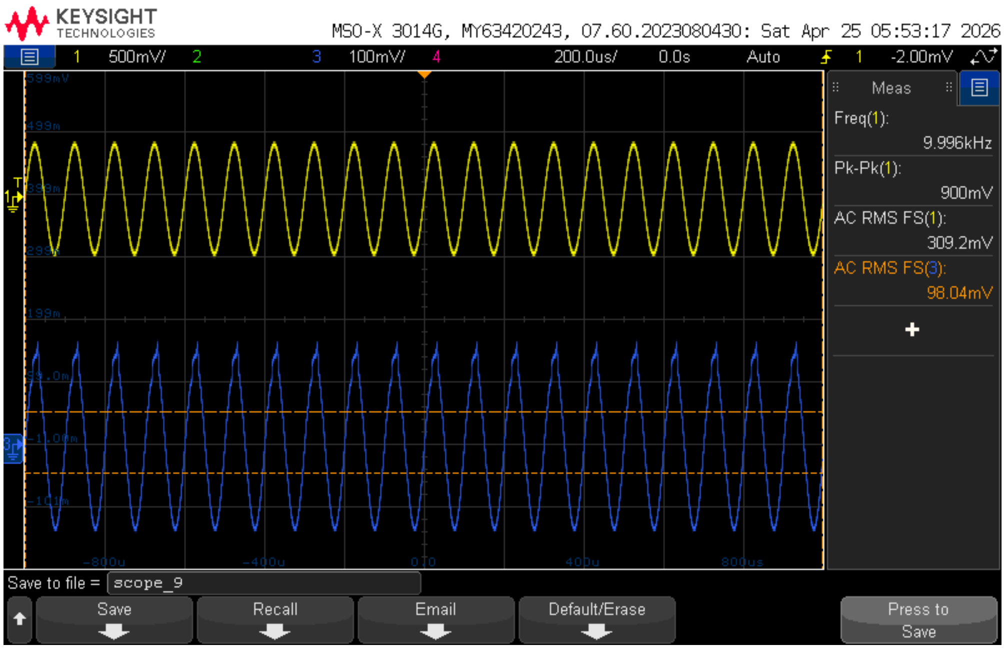
FRA:




Maximum Ripple Voltage:



Minimum Ripple Voltage:




<u>Explanation:</u>

- FRA: This shows the overall system gain across the 100 Hz to 10 kHz sweep. The blue curve shows the ripple, and cursor 1 shows the peak gain (approximately -9.06 dB at 1.259 kHz), and cursor 2 shows the minimum gain within the band at approximately (-10.15 dB at 10 kHz).
- Maximum Ripple Voltage: At the 1.259 kHz peak identified by the FRA, the measured output voltage reached its highest point of 108.91 mVrms.
- Minimum Ripple Voltage: At the 10 kHz point from the FRA, the measured output voltage dipped to its lowest point of 98.04 mVrms.

## 4.7 Ripple calculations

The ripple was calculated using the absolute voltage difference between the maximum and minimum AC RMS output voltages when a constant input signal was applied.

Measured Values:

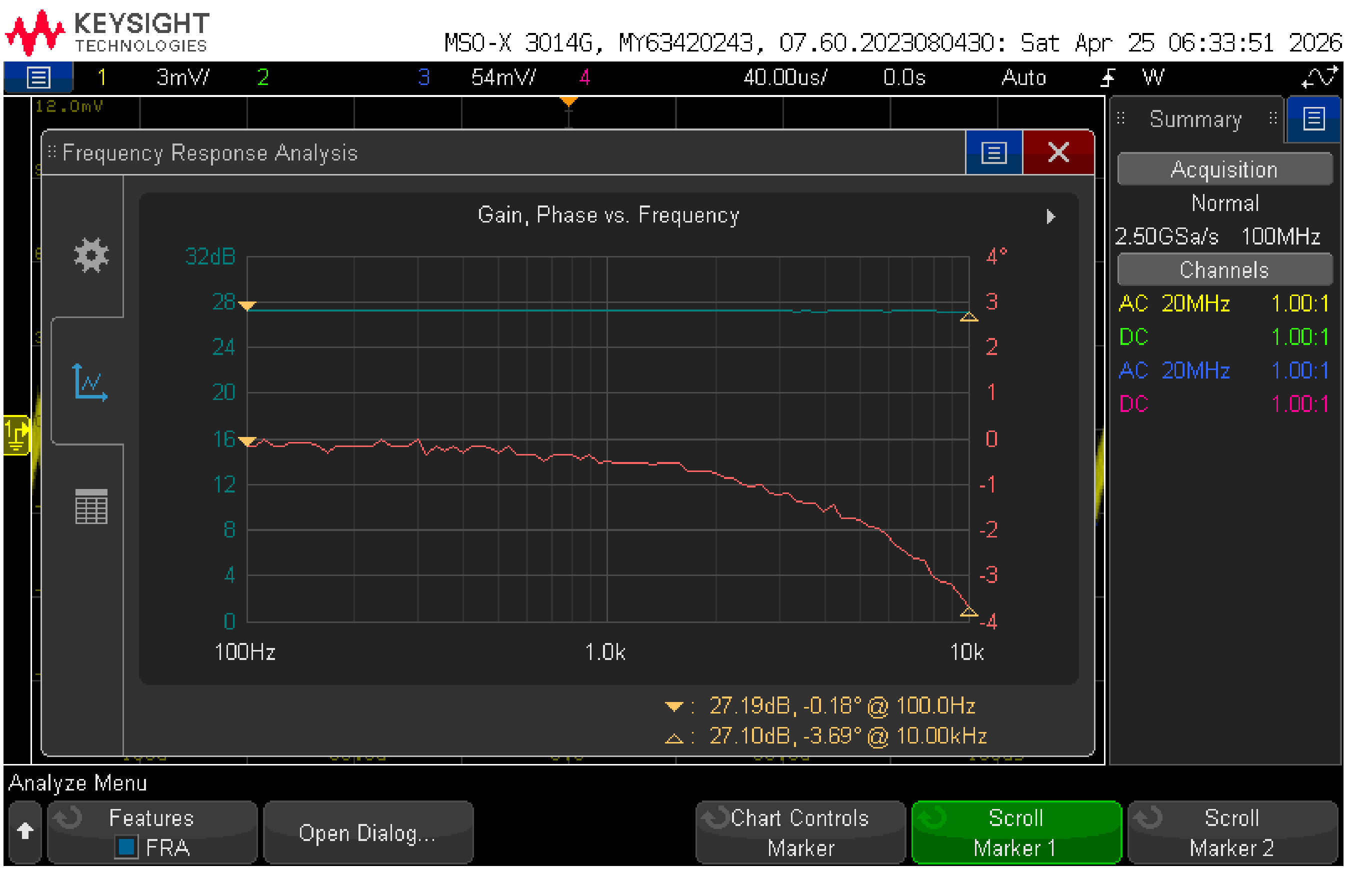
- Max passband voltage ($V_{max}$) = 108.91 mVrms at 1.259 kHz
- Min passband voltage ($V_{min}$) = 98.04 mVrms at 10 kHz

Ripple Calculation:

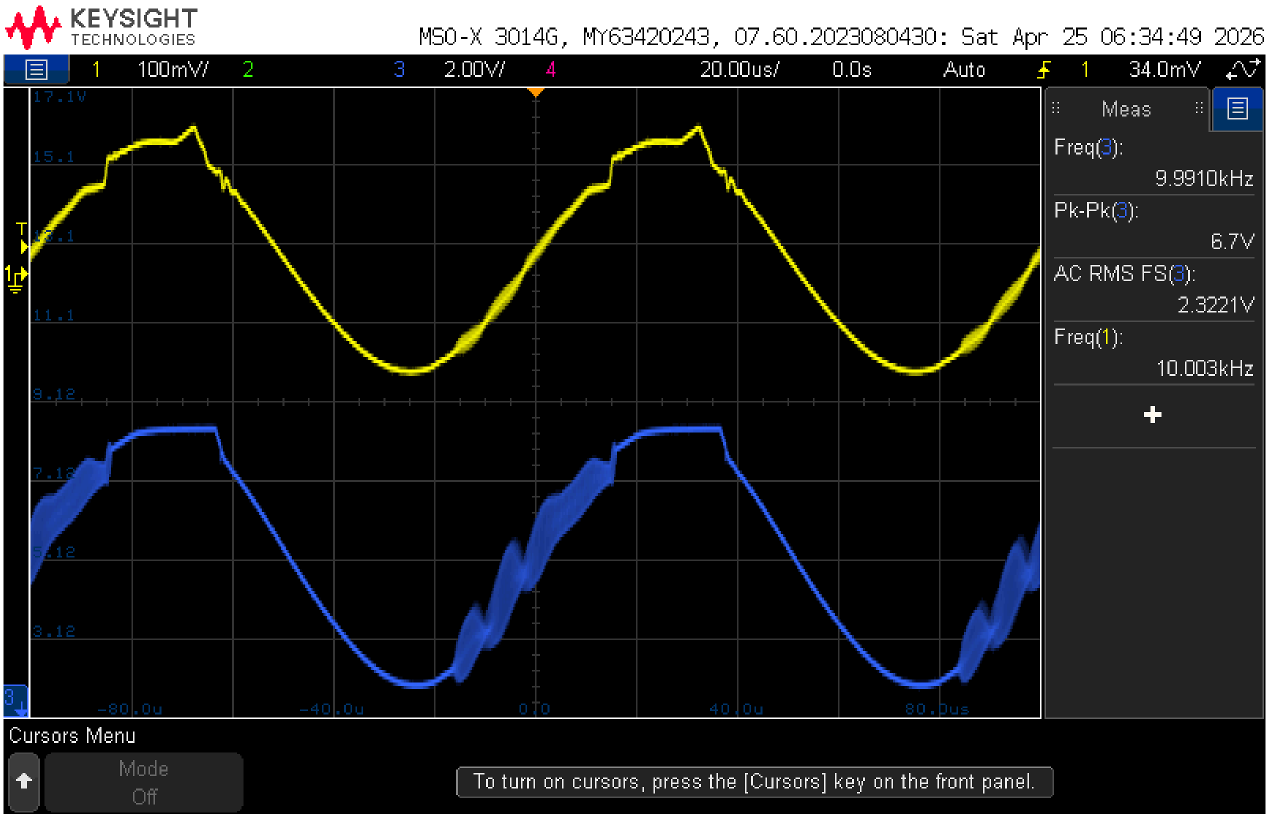
- $Ripple = V_{max} - V_{min}$
- $Ripple = 108.91\ mVrms - 98.04\ mVrms = 10.87\ mVrms$

**4.8 Plots showing Power Amplifiers Output voltage to satisfy required conditions**

FRA:



Oscilloscope Measurement:




**<u>4.9 Calculation of output power from measured output voltage</u>**

The physical power output can be calculated using the standard equation: $P = \frac{V_{rms}^{2}}{R}$

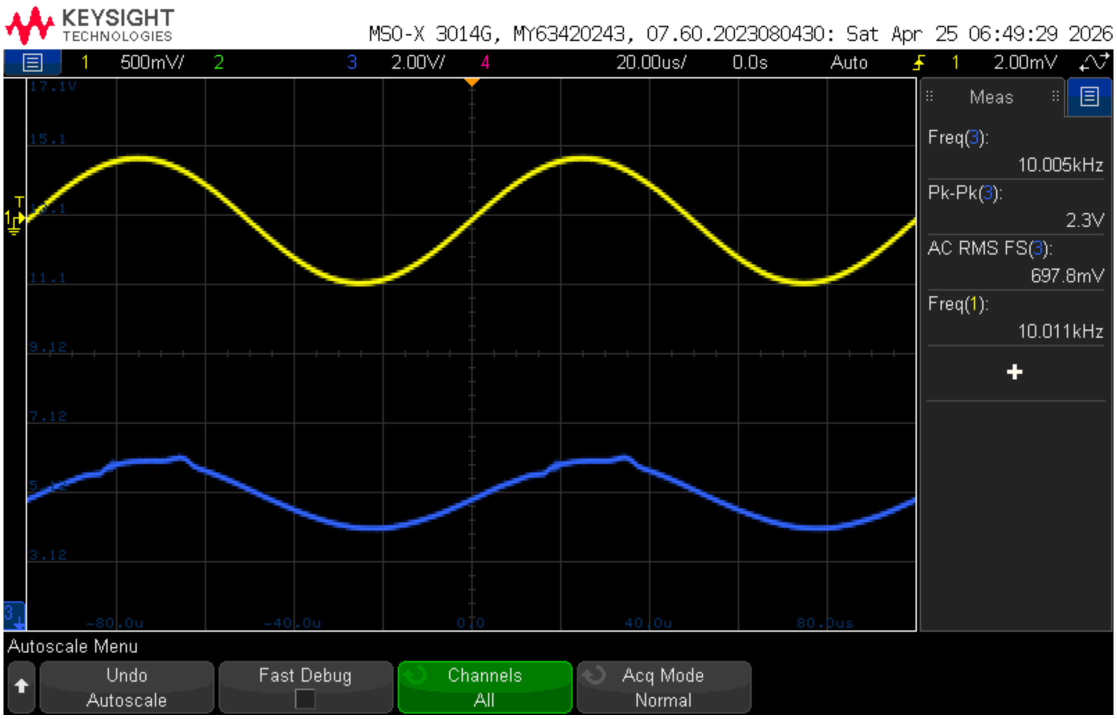
Using the measured voltage (2.3221 Vrms) and the physical speaker resistance (8 ohms):

$$P = \frac{2.3221^2}{8} = 0.674 = 674mW > 400mW$$

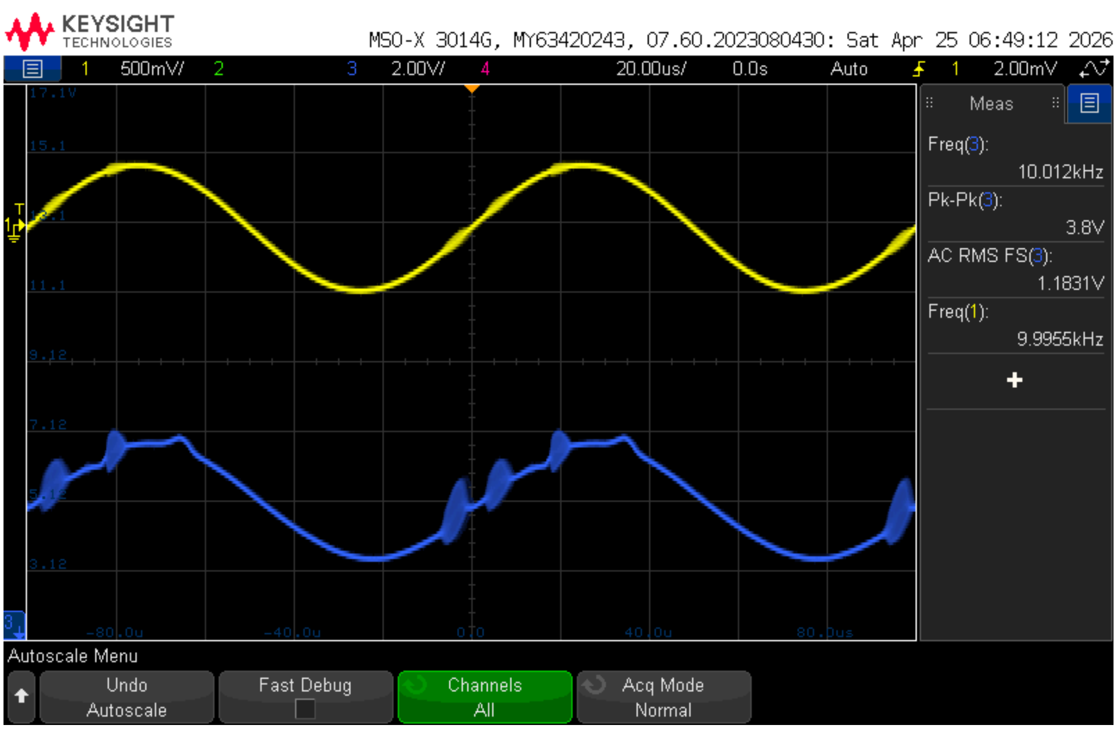
The calculated power output is 674 mW which successfully proves the LM386 power amplifier stage satisfies the > 400mW design requirement.

## 4.10 Plots demonstrating the low, medium and high-volume control

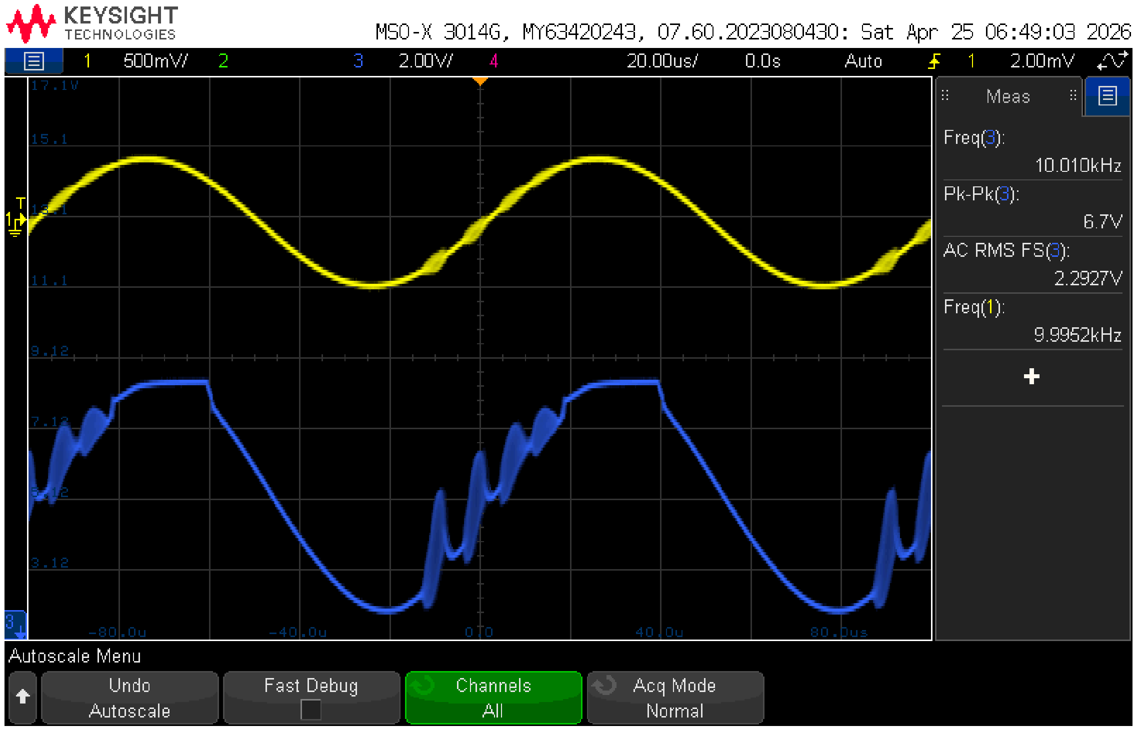
Low Volume:




Medium Volume:



High Volume:




## 4.11 Results justification, explanation, error/ tolerance calculation

**Filter Cutoff Tolerances and Results:**

- Bass (LPF): Target 320 Hz, Measured 344.53 Hz (7.66% error)
- Mid (BPF): Target 320 Hz / 3200 Hz, Measured 341.74 Hz / 3305.26 Hz (6.79% / 3.29% error)
- Treble (HPF): Target 3200 Hz, Measured 3407.02 Hz (6.47% error)

**Gain and Recombination Tolerances and Results:**

- 100 Hz Output: 103.07 mVrms (3.07% error)
- 1 kHz Output: 106.63 mVrms (6.63% error)
- 10 kHz Output: 100.62 mVrms (0.62% error)

Additionally, the minimum voltage settings successfully cut the signals down to the microvolt range which satisfies the below 15 mVrms requirement.

**Ripple Justification and Results:**

The target specification for the passband ripple was 15mVrms with a 10% tolerance, but the physical circuit measured a ripple of 10.87 mVrms. This measurement falls below the target window, but having lower ripple could possibly be from the system actually over performing since lower ripple resulted in a flatter more ideal overall frequency response. One cause for this may be from the wide tolerances of the kit components that were used.

**Power Delivery Justification and Results:**

The power amplifier stage was required to deliver > 400 mW to an 8 ohm speaker load. The LM386 successfully outputted 2.3221 Vrms, which calculates to an actual power output of 674 mW. This comfortably meets the requirement and ensures sufficient current delivery to drive the physical speaker.

## Conclusion and reference:

### 5.1 Conclusion

The main idea of this project was to design, construct, and verify an active audio equalizer capable of splitting a full-range audio signal into three bands (bass, mid, and treble), applying independent volume control to each, and then amplifying them into a recombined signal to drive an 8 ohm speaker. Overall, the physical circuit fulfilled the design requirements. The LM324 active filters properly worked at the 320 Hz and 3200 Hz points, the inverting amplifier properly allowed for the manual attenuation from maximum volume down to microvolt levels, and the LM386 power amplifier successfully delivered 674 mW of power to the speaker, exceeding the 400 mW requirements.

While the circuit operated successfully, the experimental results did have slight deviations from the ideal theoretical calculations. Across all the filter stages, the measured cutoff frequencies were slightly higher than the 320 Hz and 3200 Hz targets, resulting in percent errors ranging from 3.29 to 7.66 %. Similarly, the recombined output voltage at max volume was 6.63% above the 100 mVrms target, and finally the ripple was measured at 10.87 mVrms which was lower than the expected 15 mVrms target. A possible reason for the differences in the measurements could primarily be due to the component tolerances, since standard capacitors and resistors rarely match their listed values exactly.

## 5.2 References

Works Cited (IEEE):

[1] "Active Low Pass Filter," *Basic Electronics Tutorials*. https://www.electronics-tutorials.ws/filter/filter_5.html (accessed Apr. 26, 2026).

[2] "Active High Pass Filter," *Basic Electronics Tutorials*. https://www.electronics-tutorials.ws/filter/filter_6.html (accessed Apr. 26, 2026).

[3] "Active Band Pass Filter," *Basic Electronics Tutorials*. https://www.electronics-tutorials.ws/filter/filter_7.html (accessed Apr. 26, 2026).

[4] "Inverting Operational Amplifier," *Basic Electronics Tutorials*. https://www.electronics-tutorials.ws/opamp/opamp_2.html (accessed Apr. 26, 2026).

[5] "Non-inverting Operational Amplifier," *Basic Electronics Tutorials*. https://www.electronics-tutorials.ws/opamp/opamp_3.html (accessed Apr. 26, 2026).

[6] "Summing Amplifier," *Basic Electronics Tutorials*.

https://www.electronics-tutorials.ws/opamp/opamp_4.html (accessed Apr. 26, 2026).